

TET4565- Electricity Markets and Energy System Planning, Specialisation Course

Project assignment

Power Market Optimisation Under Uncertainty, Decomposition and Hydropower Planning Module

In this project, you will develop and analyse a stochastic optimisation model through three progressively more advanced stages. You are encouraged to choose a problem formulation of your own. This may be a variant of the problem studied in your project thesis or another topic that interests your group. The same core problem should, as far as possible, be carried through all three parts so that each stage builds on the previous one.

The project shall be completed in groups of 2–3 students. All group members must be able to explain the problem formulation, analysis, results, and code. The teaching team may ask individual students to explain parts of the submission in order to verify individual understanding and contribution.

The problem should be relevant to energy markets, production planning, flexibility in grid operations, or power-system expansion. The inspirational reading list later in this document may help you identify a suitable topic. To ensure that the problem is appropriate for all three parts of the project, it should satisfy the following requirements:

- Uncertainty must be present and must affect at least some decisions or outcomes.
- The optimisation problem must be linear, either inherently or through a clearly explained linear approximation.
- The problem must include recourse: after some uncertain information becomes available, at least one decision can be adjusted.
- The problem should preferably have only one main state/linking variable per time step, or otherwise remain simple enough for the decomposition task in Part 3.

If your group does not identify a suitable problem, you may use the short-term hydropower scheduling problem presented at the end of this document.

Project structure and progression

The project is intentionally scaffolded. Each part reuses and extends the work from the previous part:

Part	Focus	Main outcome	Points
1	Problem formulation and data	A well-motivated two-stage stochastic model and a prepared dataset	25
2	Model implementation and comparison	Expected-value, scenario-by-scenario, and stochastic solutions that can be compared	30
3	Benders' decomposition	A decomposed implementation validated against the Part 2 stochastic solution	45

Part 1: Problem formulation and data cleaning (25 points)

Formulate your chosen problem as a two-stage stochastic optimisation problem and carry out a focused data-analysis exercise. The purpose of this part is to establish a clear decision problem, uncertainty structure, mathematical formulation, and data basis before implementation begins.

Learning goals:

- Explain the motivation and practical value of the decision problem, and position it relative to relevant literature or the research frontier.
- Formulate a two-stage decision problem under uncertainty mathematically.
- Identify the relevant uncertain variables and information structure, including what is known when each decision is made.
- Process, clean, transform, visualise, and describe relevant data, preferably using Pandas in Python.

Tasks:

- Briefly describe the optimisation problem that you aim to solve. Your description should make the following elements explicit:
 - o Context: What background information is needed to understand the problem?
 - o Decision problem: Describe the problem intuitively in words.
 - o Value and stakeholders: Why is the problem worth solving, and who is affected by the decisions?
 - o Uncertainty and information: Identify the uncertain variables, when they become known, and which decisions are made before and after that information is available.
- Research question: Summarise the purpose of the model as one or a few explicit research questions.
- Mathematical formulation: Write the optimisation problem in mathematical form, with clearly defined notation, objective function, constraints, and stage-specific decisions.

Working with a dataset that is directly relevant to your problem is encouraged. Because this is early in the semester, you may instead use one of the following options:

- Your own dataset
- The Rye Microgrid dataset (Dataset A in this folder)
- The Austin Household consumption dataset (Dataset B in this folder)

The two provided datasets include explicit questions. If you work with your own dataset, your analysis should at minimum address the following:

- Read and present the data using relevant plots and/or summary statistics.
- Identify missing values and outliers, and explain how you handle them.
- Aggregate or transform the data where appropriate, for example by resampling time-series data or grouping observations.
- Identify relevant patterns in individual series and, where useful, correlations between series.

Deliverables:

- A concise written formulation of the optimisation problem, including motivation, value, decisions, and uncertainty structure (maximum 1 page).
- One or a few research questions that the model will be used to answer.
- A mathematical formulation of the optimisation problem, including notation.
- Code used for data handling and analysis.
- A brief, integrated description of the data and the main observations from the analysis (maximum 1–2 pages, including plots). Do not answer dataset questions as isolated numbered bullets; instead, write a coherent summary that incorporates the answers.

The data description and data-handling code may be combined in a Jupyter notebook if preferred.

Part 2: Programming and comparing stochastic formulations (30 points)

Implement and solve the model computationally. Python with Pyomo is recommended. You may use an available open-source solver or an academic Gurobi licence. The emphasis is on a transparent implementation that makes the effect of uncertainty on decisions easy to analyse.

If some model inputs are not yet available, you may use transparent synthetic data. For uncertain inputs, simple low/medium/high scenarios or draws from a random variable are acceptable. If the scenario set becomes large, you may reduce it using a clearly explained method.

Solve and compare the following three variants of the same problem:

- Expected-value approximation: replace uncertain inputs by their expected values and solve the resulting deterministic problem.
- Scenario-by-scenario solutions: solve the problem separately for each scenario, assuming that scenario is known in advance.
- Stochastic solution: solve all scenarios jointly while enforcing nonanticipativity for first-stage decisions.

Learning goals:

- Implement working, readable, and well-structured optimisation code.
- Explain the conceptual and numerical differences between the expected-value approximation, scenario-by-scenario solutions, and the stochastic solution.

Tasks:

- Write code that solves all three variants consistently using the same underlying problem and data.
- Present the key decisions and objective values for all three variants, and compare them using tables and/or figures where helpful.

Deliverables:

- A concise presentation and discussion of the results, focusing on the differences between the three variants and why those differences arise (maximum 2 pages).
 - Code for all three model variants.
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Part 3: Benders' decomposition (45 points)

In Part 2, you solved the stochastic problem as a deterministic equivalent. As the number of scenarios and the size of the model increase, the deterministic equivalent can become large. In this part, you will decompose the two-stage stochastic problem using Benders' decomposition. The goal is both to obtain the correct solution and to understand how the master problem, scenario subproblems, cuts, and convergence bounds interact.

Learning goals:

- Identify the first-stage decisions, second-stage recourse decisions, and the constraints that link the two stages.
- Structure the implementation using reusable functions that parameterise, build, and solve the master problem and subproblems.
- Use dual information to formulate valid Benders cuts and interpret the convergence process.

Tasks:

- Implement Benders' decomposition with a master problem and scenario subproblems, and define valid upper and lower bounds for measuring convergence.
- For each scenario, obtain the dual multiplier(s) associated with the constraint(s) linking the subproblem to the fixed first-stage decision(s). Use these dual values to construct the scenario contribution to a Benders cut. Combine scenario contributions using their probabilities; for equiprobable scenarios, this is a simple average.
- Iterate until the gap between the upper and lower bounds satisfies a clearly stated convergence tolerance.
- Compare the converged objective value and first-stage decisions with the stochastic deterministic-equivalent solution from Part 2. Explain any small numerical differences.

Deliverables:

- A mathematical formulation of the decomposed problem, including the master problem, subproblem(s), bounds, and Benders cut.
 - Code that implements the decomposition described above.
 - A concise convergence summary and comparison with the Part 2 stochastic solution, supported by a small table or figure where useful.
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Assessment and grading rubric

The project is graded out of 100 points. The rubric below mirrors the three-part project structure and should be used as a checklist while you work. The final score is the sum of the points awarded across Parts 1–3, less any applicable report-length/format deduction.

Part	Assessment criterion	Points
Part I	Context and motivation	2
Part I	Problem description	3
Part I	Decision identification	1
Part I	Uncertainty description	3
Part I	Research question	2
Part I	Optimisation problem (mathematical form)	7
Part I	Data analysis	7
	SUBTOTAL PART I	25
Part II	Expected value of all scenarios	4
Part II	Each scenario individually	4
Part II	Stochastic problem	9
Part II	Analysis, comparison, and explanation	9
Part II	Code quality	4
	SUBTOTAL PART II	30
Part III	Identification of first/second stages and linking	6
Part III	Mathematical formulation	7
Part III	Definition of upper and lower bounds	6
Part III	Identification of the dual variable(s)	6
Part III	Formulation of the cuts	10
Part III	Comparison with Part II	6
Part III	Code quality	4
	SUBTOTAL PART III	45

Report length/format: 1-point deduction if the report exceeds 13 pages.

Grade thresholds

Grade	Total points
A	89–100
B	77–88
C	65–76
D	53–64

E	41–52
F	0–40

Before submission, check that: the decision and uncertainty structure are explicit; all mathematical symbols are defined; the Part 2 variants are directly comparable; the Benders solution is validated against Part 2; code is readable and reproducible; and every group member can explain the complete submission.

Inspirational reading list

You are by no means required to read any of the paper below, but some of them might give you ideas for a problem formulation similar to the one you want to work with.

1. T Trötscher, M Korpås, "A framework to determine optimal offshore grid structures for wind power integration and power exchange", *Wind Energy* 14 (8), 977-992
2. H. Farahmand, T. Aigner, G. L. Doorman, M. Korpas and D. Huertas-Hernando, "Balancing Market Integration in the Northern European Continent: A 2030 Case Study," in *IEEE Transactions on Sustainable Energy*, vol. 3, no. 4, pp. 918-930, Oct. 2012
3. Ari Hentunen, Ville Erkkilä, Samppa Jenu, " Storage system dimensioning and design tool", D6.1, INVADE H2020 project
4. Espen Flo Bødal, Pedro Crespo del Granado, Hossein Farahmand, Magnus Korpås, Pol Olivella, Ingrid Munné and Pau Lloret , "Challenges in distribution grid with high penetration of renewables", D5.1, INVADE H2020 project
5. M. Korpås and A. T. Holen, "Operation planning of hydrogen storage connected to wind power operating in a power market," in *IEEE Transactions on Energy Conversion*, vol. 21, no. 3, pp. 742-749, Sept. 2006, doi: 10.1109/TEC.2006.878245.
6. Thorsten Burandt, Bobby Xiong, Konstantin Löffler, Pao-Yu Oei," Decarbonizing China's energy system – Modeling the transformation of the electricity, transportation, heat, and industrial sectors", *Applied Energy*, Volume 255, 2019
7. Ove Wolfgang, Arne Haugstad, Birger Mo, Anders Gjelsvik, Ivar Wangensteen, Gerard Doorman," Hydro reservoir handling in Norway before and after deregulation", *Energy*, Volume 34, Issue 10, 2009, Pages 1642-1651,
8. Kong Jiehong, Skjelbred Hans Ivar, Fosso Olav Bjarte. "An overview on formulations and optimization methods for the unit-based short-term hydro scheduling problem". *Electr Power Syst*
9. Gjelsvik A., Mo B., Haugstad A. (2010) Long- and Medium-term Operations Planning and Stochastic Modelling in Hydro-dominated Power Systems Based on Stochastic Dual Dynamic Programming. In: Pardalos P., Rebennack S., Pereira M., Iliadis N. (eds) *Handbook of Power Systems I. Energy Systems*. Springer, Berlin, Heidelberg. http://www.newbooks-services.de/MediaFiles/Texts/4/9783642024924_Excerpt_001.pdf
10. Julija Matevosyan, Magnus Olsson, Lennart Söder, "Hydropower planning coordinated with wind power in areas with congestion problems for trading on the spot and the regulating market", *Electric Power Systems Research*, Volume 79, Issue 1, 2009, Pages 39-48

11. Helseth, Arild; Mo, Birger; Lote Henden, Arild; Warland, Geir: "Detailed long-term hydro-thermal scheduling for expansion planning in the Nordic power system", IET Generation, Transmission & Distribution, 2018, 12, (2), p. 441-447, DOI: 10.1049/iet-gtd.2017.0903 IET Digital Library, <https://digital-library.theiet.org/content/journals/10.1049/iet-gtd.2017.0903>
 12. Farahmand, Hossein; Jaehnert, Stefan; Aigner, Tobias; Huertas-Hernando, Daniel, "Nordic hydropower flexibility and transmission expansion to support integration of North European wind power", Wind Energy 2015 ;Volume 18.(6) s. 1075-1103
 13. Kasper Emil Thorvaldsen, Magnus Korpås, Karen Byskov Lindberg, Hossein Farahmand, "A stochastic operational planning model for a zero emission building with emission compensation", Applied Energy, Volume 302, 2021
 14. Widén, J., Lundh, M., Vassileva, I., Dahlquist, E., Ellegård, K., & Wäckelgård, E. (2009). Constructing load profiles for household electricity and hot water from time-use data- Modelling approach and validation. Energy and Buildings, 41(7), 753-768. <https://doi.org/10.1016/j.enbuild.2009.02.013>
 15. Sperstad IB, Korpås M. Energy "Storage Scheduling in Distribution Systems Considering Wind and Photovoltaic Generation Uncertainties". *Energies*. 2019; 12(7):1231. <https://doi.org/10.3390/en12071231>
 16. Stein-Erik Fleten, Trine Krogh Kristoffersen "Stochastic programming for optimizing bidding strategies of a Nordic hydropower producer", European Journal of Operational Research, 2007, 181(2), p 916-928, <https://doi.org/10.1016/j.ejor.2006.08.023>
 17. JM Morales, M Zugno, S Pineda, P Pinson [Electricity market clearing with improved scheduling of stochastic production](#), European Journal of Operational Research, 2014
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Short-term hydropower optimisation problem

This is a problem you can work on if you can't think of another suitable problem for these assignments.

You, as the newest hydropower-station employee, are asked to determine the optimal production strategy for the next 48 hours using forecasts of inflow and electricity price. The first 24 hours are assumed to be known, while five scenarios are provided for the remaining 24 hours. In a two-stage formulation, the common decisions for the first 24 hours are made before the second-stage scenario is known, while the decisions for hours 25–48 may adapt to the realised scenario. Your model should therefore balance the expected consequences of all five scenarios when determining the first-stage strategy.

The following information is given about the hydropower unit:

Parameter	Value	Information
V_0	3 [Mm3]	The starting reservoir level for the hydropower station
V_MAX	4.5 [Mm3]	Rated capacity for the reservoir
Q_Max	100 [m ³ /s]	Maximum discharge capacity of water from the reservoir to the hydropower unit
P_MAX	86.5 [MW]	Maximum production capacity for the hydropower unit
M3S_TO_MM3	3.6/1000 [Mm3/m ³]	Conversion factor from cubic meter per second to Mm3 (million cubic meter)
E_conv	0.657 [kWh/m ³]	Power equivalent from discharged water to produced electricity.
WV_end	52600 [EUR/Mm3]	Water value for leftover hydropower at the end of the 48 th hour.

The following information is given about the inflow and prices for the next 48 hours:

Parameter	Value	Information
Inflow [First 24 hours]	25 [m ³ /s]	The inflow for the first 24 hours, given hourly.

Inflow [Last 24 hours]	$10 \cdot s$ [m^3/s]	The inflow for the last 24 hours, given hourly. This includes uncertainty and a 0-index formulation. Example: For scenario 0, inflow is $10 \cdot 0 = 0$
N_Scenarios	5 [-]	The number of scenarios for the second stage (last 24 hours)
ρ_{scenario}	1/5 [per unit]	The probability for each scenario.
Price	$50 + t$ [EUR/MWh]	The power prices for all 48 hours, given as a linearly increasing cost based on time step t . Assumes 0-index. Example: For hour 13 the cost is $50 + 13 = 63$.